

The Car That Cried Wolf: Driver Responses to Missing, Perfectly Performing, and Oversensitive Collision Avoidance Systems

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Abstract— Automated emergency braking (AEB) systems—which alert a driver to approaching hazards and automatically brake—are currently available in some vehicles and will soon be widespread. Due to the uncertainties inherent in any environment and difficulties in processing sensor data, these systems are prone to both false alarms and system misses of hazards. A pressing design concern is whether to bias these systems toward a higher likelihood of false alarms versus system misses for non-fatal events. We investigated how drivers form mental models of the AEB system and how that influences their reliance on the system during a critical, potentially fatal, failure. In a full vehicle driving simulator, participants experienced nine interactions that reflected the system’s level of bias toward false alarms or toward misses for non-fatal events or that demonstrated perfect performance. When a potentially fatal event occurred, participants trained to expect misses were better able to avoid a pedestrian in the road after a detection failure than those using a system with perfect performance or false alarms. These findings suggest that systems biased toward misses in non-fatal events encourage driver vigilance and preparedness for potentially fatal events.

I. INTRODUCTION

Active safety features, such as automated emergency braking (AEB), constitute lifesaving technology that is quickly becoming prevalent in passenger vehicles. Because AEB provides an alert and automated braking when a hazard is detected ahead of a vehicle and a collision is imminent, this feature improves driver performance by both providing a timely alert and boosting brake force if necessary. AEB can also prevent or mitigate a collision independent of driver action by activating the brakes if the driver is inattentive or impaired [1].

However, AEB can pose a potential hazard if the system brakes even though no threat exists ahead of a vehicle. In such a case, a driver needs to override the false alarm, which could be difficult and dangerous depending upon the road conditions and the design of the system. Moreover, another vehicle could be following closely behind, and unnecessary braking could cause a rear-end collision [2].

Given this potential hazard, most, if not all, AEB systems are configured to wait until a collision is nearly unavoidable to initiate braking, thereby giving the driver ample time to initiate a braking response before a collision [3]; this design choice has the added benefit of not training the driver to rely on the AEB system. However, if a driver does rely on the system, and it fails to detect an obstacle or does not brake in



Figure 1. The Stanford Driving Simulator

response, the driver may be unable to recover from such a failure due to the limited time remaining. Configuring the sensitivity of an AEB system is thus a crucial design consideration.

This study explored the relative merits of various sensitivity settings for an AEB system, comparing a system that misses hazards for non-fatal events (the pedestrians in our simulation were not struck) to a system that is overly sensitive and delivers false alarms. Both of these systems were compared against a perfectly performing system that correctly rejects noise and detects hazards. Over the course of nine training events, drivers were able to build mental models of the system’s capability, which was tested empirically when the system failed to detect a potentially fatal hazard, or did not brake to avoid the hazard and provided only an audible alert.

Initial analysis of the notification system, conducted in a full-vehicle fixed-base driving simulator (see Fig. 1), indicates that a system which includes occasional misses of non-fatal hazards may lead to better driver performance when an eventual critical, potentially fatal, event does occur.

II. BACKGROUND

When using any system, a person creates a mental representation, or mental model, of the system’s capabilities and how to operate it. Experience helps build a more accurate model in the user’s mind, which is more easily called upon in use [4, 5, 6]. With an Advanced Driver Assist System (ADAS) such as the one simulated in this study, an accurate and usable mental model is of great importance in teaching drivers how to properly rely on the system. Here we relied heavily on the mental model participants constructed through experience with the system, which they then applied to a novel situation—the final event. We based the system and mental model on the capabilities of existing ADAS systems, along with current knowledge of trust and reliance in automation, and the effect of varying levels of alerts in driver interactions with warning systems.

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A. Advanced Driver Assist Systems (ADAS)

Automated emergency braking is a type of ADAS that can prevent frontal collisions or significantly reduce collision energy if one is unavoidable. The feature is increasingly common in passenger vehicles and will be a mandatory standard feature in the US starting in 2022 [7]. Aside from the technical issues of creating a system that properly detects hazards and rejects noise, designers must make judgments about whether to bias the system to be more sensitive and risk more false alarms, or to decrease sensitivity and risk more misses.

Many manufacturers have implemented AEB using RADAR, visual spectrum cameras, or a combination of both. Each technology can greatly assist drivers but also has its drawbacks relating to expense, sensing, or signal processing [8], which can lead to false alarms when there is no threat of collision [9]. Sensor fusion, which combines the inputs from multiple sensing systems, is being actively developed [10], but no system, regardless of how sophisticated, will be entirely impervious to false alarms and misses.

An additional limitation is the common inability of current systems to sense *behind* the vehicle (hence the term forward collision warning); in the case of a braking activation, such systems may cause a rear-end collision if a vehicle is following too closely [11]. If there is a hazard ahead, this limitation may still be preferable to striking an obstacle, animal, pedestrian, or cyclist, but, in the case of a false alarm, this could be disastrous.

When there is a hazard ahead, systems are configured to provide an alert, and then actuate the brakes if the driver's response is insufficient—either too late, or not strong enough [12]. This dual-staging allows for an incomplete failure of the system, where the system provides an alert, but does not actuate the brakes. Braking systems, when combined with the collision warning capabilities, thus present an area that could benefit from further research.

B. Signal Detection and Discrimination

Signal Detection Theory describes discrimination between true signals and noise, classifying signals and reactions to them. A signal can be present (signal and noise) or absent (only noise), and there can be a response, or no response [13, 14]. This is represented by the 2x2 matrix in Table 1.

TABLE I. SIGNAL DETECTION THEORY

	Signal Absent (noise)	Signal Present (noise + signal)
No Detection	Correct Rejection	Miss
Detection	False Alarm	Correct Detection (hit)

Signal detection theory is based on the truth of the signal (present or absent) and the response to it.

A great deal of research has been conducted on human signal detection and classification, and there is a corresponding large body of literature on machine sensing and classification [15]. Human and machine capabilities are tandem problems in the situation we investigate here—a system needs to properly detect and classify signals and noise, and the human driver likewise needs to do so.

With a driver assistance system, the computational difficulty is high due to the diversity of items in the

environment—e.g., other vehicles, cyclists, pedestrians, animals, or debris in the roadway—and the system needs to react appropriately to all of them. That is, the computer needs to properly classify a small child and correctly reject the reflection from a plastic bag blown by the wind. And, although humans are very good at classifying such things, they are not capable of sustained vigilance, even under ideal circumstances [16, 17, 18].

In the construction of a robust and effective system that fosters driver vigilance, a criterion level must be selected to divide signals from noise, and this problem motivates our research—should the system be biased towards greater sensitivity and accept more false alarms in exchange for fewer misses, or towards lower sensitivity, accepting more misses in exchange for fewer false alarms? This tradeoff is generally determined by the relative cost of misses and false alarms—in this case, the potential failure to prevent or mitigate a collision, or the risk of causing one.

C. Trust and Reliance in Automation

Drivers need to trust and rely on a vehicle's automated system to be willing to use it. Mayer et al. describe trust that can be applied to studying automation as “The willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor.” They further consider risk to be an inherent part of the trust relationship, meaning that there has to be a possibility of being let down [19]. In this study, the risk analyzed in the automated system is that a system failure could lead to a collision: a miss could cause the participant to strike a pedestrian, or a false alarm could cause a rear-end collision if the vehicle behind is following closely.

Participants' mental models of the system's capability are intertwined with their perception of its reliability and trustworthiness, and their preexisting trust in technology. This trait can be measured to some degree by a questionnaire (here, we used Jian, Bisantz, and Drury's instrument [20]). Use of the system can also change participants' level of trust, which can then be compared with their reliance on it while driving.

Overtrust in automated driving features has led to several accidents that have received a great deal of media attention [21], leading to increased scrutiny of the design of such systems, and highlighting drivers' inability to maintain effective supervisory control for protracted periods [22].

Lee and See discuss the concept of appropriate reliance, where one should rely only on a system that is sufficiently trustworthy and, ideally, has demonstrated that it can function adequately. This requires a properly “calibrated trust” that is matched to the system's capabilities and therefore justifies reliance. This calibration derives from the driver's mental model of the system's capability, which is itself formed by interaction with the system [23]. Consequently, the more experience the user has with the system, and the more diverse the situations are, the better the calibration will be. In a simulation study, only a few experiences are possible due to limited participant time and researchers' desire to reflect real-world settings. In the real world, a sophisticated system such as AEB will provide a training stimulus whenever it is activated, but that activation may be infrequent, as these safety systems are (fortunately) rarely needed. Thus, drivers

will be confronted with system activation in difficult, time-limited situations with relatively minimal prior experience, making design of the system critical to maximizing the correct driver behavior in these situations.

D. Complacency and Nuisance with Alerts

Varying levels of alerts with warning systems can result in both operator complacency and nuisance, a response that has been assessed in aviation studies with pilots [24, 25, 26].

Complacency can occur when pilots are so dependent on the alert system that they fail to pay adequate attention and vigilance is reduced; that is, they rely completely on the system. However, Wiener provides another interpretation of this behavior, describing it as primary-secondary task inversion, where the primary task of paying attention is instead substituted with the secondary task of simply listening for the alerts and alarms that indicate warnings [27].

Nuisance, i.e. when a system has too many false alarms, can lead to an alternative problem. In cases of excessive number of alerts or alarms, pilots have ignored the alerts because they think of them as unnecessary or false [28]. In addition to reducing vigilance, an excessive number of alerts can sometimes even cause overload and confusion [29].

While alert frequency has been studied with warning systems in the aviation context, it is important to study these effects in automotive settings. The road setting requires much faster response time than aviation: a driver may have a fraction of a second to respond before a fatal incident, while a pilot may have minutes or more, depending on the cruising altitude and speed of an aircraft.

III. STUDY GOALS AND METHODS

Hypothesis: We hypothesized that if an autonomous system fails to detect hazards, drivers will increase their vigilance to compensate for the system's poor performance. Alternatively, if the system exhibits false alarms, drivers will decrease their vigilance—much like the townspeople in Aesop's fable who eventually ignore the shepherd boy who had falsely cried wolf too many times. Finally, with perfect performance, drivers' vigilance will likely also decrease as complacency sets in, lowering their abilities to respond to hazards that the system does not recognize and respond to.

Goals: To investigate this hypothesis, we designed a study to explore the formation and use of a mental model of system performance by varying the rate of false alarms and misses between participants, and then comparing participant performance and reliance on the system during a critical failure.

A. Participants

Participants between the age of 18 to 60 years old were recruited using flyers and emails and were compensated for their time with an Amazon gift card. Participant driving experience ranged from 2 to 44 years ($M = 16.46$ years, $Mdn = 13$ years, $SD = 11.20$ years). Participants reported driving between 0.2 and 5.1 days per week ($M = 4.49$ days, $Mdn = 5.1$ days, $SD = 2.30$ days).

B. Experiment Design

Participants were assigned to one of three levels of system capability: over nine pedestrian encounters, the system would exhibit false alarms, misses, or perfect

performance. In the final, tenth event, participants would experience either a detection failure or a braking failure (see Table 2). This design provided for three pairwise comparisons: between missing and perfect performance with detection failure, between the two perfect performance conditions (distinguished by having a detection or braking failure at the end), and between false alarms and perfect performance with brake failure.

TABLE II. EXPERIMENTAL CONDITIONS

	Condition			
	Missing - Detection Failure	Perfect - Detection Failure	Perfect - Brake Failure	False Alarms - Brake Failure
Number of Participants	N = 12	N = 12	N = 12	N = 12

C. Procedure

The study was conducted in a full-vehicle cab, fixed-base driving simulator, with 270° wrap-around screen, rear view screen, and separate video channels for rear view mirrors.

Participants first drove for approximately four minutes in a suburban environment to familiarize themselves with the simulated vehicle's behavior before encountering the nine pedestrian incursion events, which were spaced over a 30-minute driving course. Participants were instructed to adhere to the speed limit of 45 mph, with two lanes in each direction. The simulated vehicle featured an automated forward collision warning system in addition to the automated emergency braking. Upon detecting a hazard, the system would provide a verbal alert "Warning: Obstacle Detected" and initiate an automated braking action.

During the simulated drive, pedestrians crossed the street in front of the vehicle six times: from the right to left four times, and from the left to right two times. Pedestrians ran along the sidewalk three times: on the right-hand side of the road two times, and on the left-hand side once. Altogether, the vehicle interacted with pedestrians ten times (see Table 3).

TABLE III. SIMULATION EVENTS AND SYSTEM PERFORMANCE BY CONDITION

	Missing - Detection Failure	Perfect - Detection Failure	Perfect - Brake Failure	False Alarms - Brake Failure
Correct Detections	3	6	6	6
Misses	3	0	0	0
False Alarms	0	0	0	3
Correct Rejections	3	3	3	0
Final Event Behavior	Detection Failure		Braking Failure	

The simulation dynamically set pedestrians' walking rate based on the vehicle's current speed to ensure that vehicle-pedestrian interactions were consistent across conditions. For example, in the exact notification condition, if the participant driving the vehicle were to slow down significantly, the

pedestrian would still walk directly in front of the car, and the car would initiate braking in time. In all but the final event, it was not possible to strike pedestrians without deliberately steering into them. We were initially concerned that participants in the condition with misses would think that the vehicle's system worked perfectly, as the vehicle did not actually hit the pedestrian crossing. However, all participants in that condition either slowed the vehicle or brought it to a full stop.

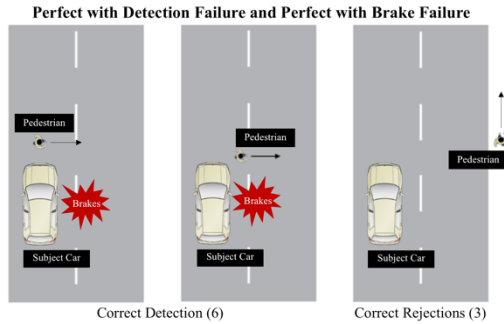


Figure 2. System exhibiting perfect performance with either detection failure or brake failure in the final event

When the system exhibited perfect performance, participants experienced the car notifying and braking during all six pedestrians crossing the street (see Fig. 2).

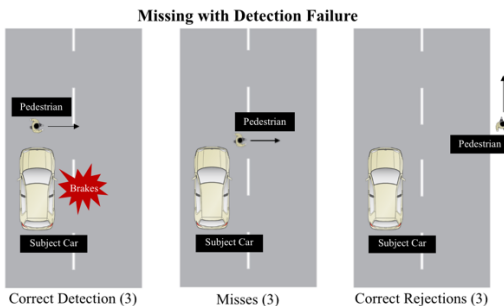


Figure 3. System exhibiting misses with detection failure in the final event

When the system exhibited misses, participants were presented with three instances where the car did not notify and brake during a pedestrian crossing (see Fig. 3). Note that in the same condition, to reflect existing industry system design, there were also some instances where the car exhibited correct performance.

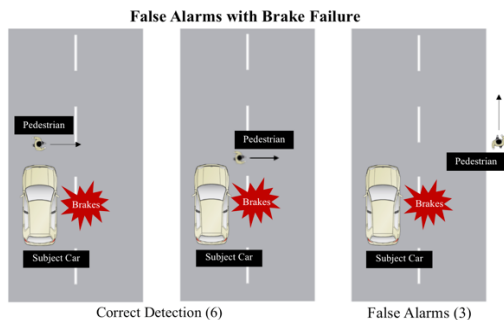


Figure 4. System exhibiting false alarms with brake failure in the final event

When the system exhibited false alarms, participants experienced notification and braking during all nine times the vehicle encountered a pedestrian, even if the pedestrian was just running along the sidewalk and not crossing the vehicle's path (see Fig. 4).

Last, all participants were presented with a tenth event, where pedestrians (a person and dog) entered the roadway at a curve and right turn (see Fig. 5). This event was used as the main metric to measure changes in driver performance between conditions.

In the tenth event, the automated braking system would either fail to detect the pedestrians (detection failure or miss), or would only provide an audible alert, without a braking system actuation (brake failure). These two failure modes provide for pairwise comparisons between the underperformance and perfect performance with detection failure condition, the two perfect performance conditions where there is a difference in failure type, and between the perfect performance with brake failure and false alarms condition.

IV. RESULTS

To analyze participant performance and reliance on the system during a critical failure, we focused our analysis on the behavior of participants in the final (tenth) event sequence. Participant behavior in the final event differed depending on variance in false alarms and misses in the preceding nine events. These results were then further analyzed with the vehicle's minimum distance to the pedestrian and driving speed, along with trust data analysis.



Figure 5. Final event, when participants must navigate a person and dog while the system exhibits failure

A. Participant Performance in Critical Event

The number of participants who navigated the final, critical event successfully was highest in the Perfect with Braking Failure condition (11). Missing and False Alarms conditions had an equal number of participants who navigated the final event successfully (10). Few participants successfully navigated the final event in the Perfect with Detection Failure condition (4) (see Fig. 6).

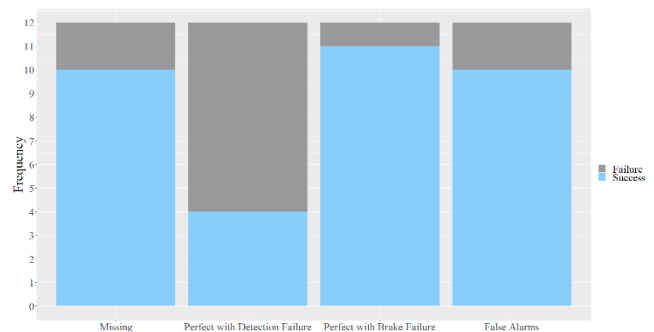


Figure 6. Performance of failure or success in navigating the final event

A Fisher's exact test showed that the proportion of participants who collided with the crossing pedestrians (the person and dog) in the final event was statistically different between the Missing (2/12) and Perfect with Detection

Failure (8/12) conditions ($p = 0.036$), calling for further analysis.

B. Minimum Distance to Pedestrian and Vehicle Speed

We calculated the vehicle's minimum distance to the pedestrians in the 6-second window following the event trigger (see Fig. 7). This duration was chosen to account for participants who drove into the pedestrians, while also accounting for participants who slowed down and later (outside of this 6-second window) cautiously drove past the pedestrians. We further examined participants' behavior and inability to navigate the final event in the Missing and Perfect with Detection Failure conditions. An independent-samples t-test to compare this minimum pedestrian distance in the Missing and Perfect with Detection Failure conditions found a marginally significant difference in the scores for Missing ($M=12.42\text{m}$, $SD=12.7\text{m}$) and Perfect with Detection Failure ($M=4.39\text{m}$, $SD=6.56\text{m}$) conditions; $t(23)=1.946$, $p = 0.069$. These results suggest that Perfect with Detection Failure participants frequently collided with the pedestrians.

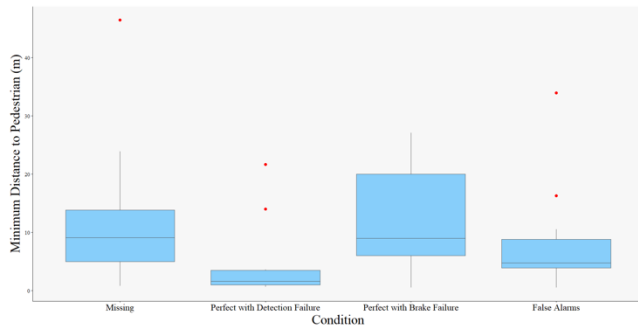


Figure 7. Minimum vehicle distance from crossing pedestrians in the final event

We also analyzed the vehicle's speed when the pedestrians were collided with (see Fig. 8). This measure captures the difference between participants who drove into or past the pedestrians at high speed and those who slowed down and drove past the pedestrians cautiously. An independent-samples t-test to compare this speed in the Missing and Perfect with Detection Failure conditions found a significant difference in the scores for Missing ($M=1.78\text{ m/s}$, $SD=2.65\text{ m/s}$) and Perfect with Detection Failure ($M=7.68\text{ m/s}$, $SD=7.1\text{ m/s}$) conditions; $t(23)=-2.696$, $p = 0.017$.

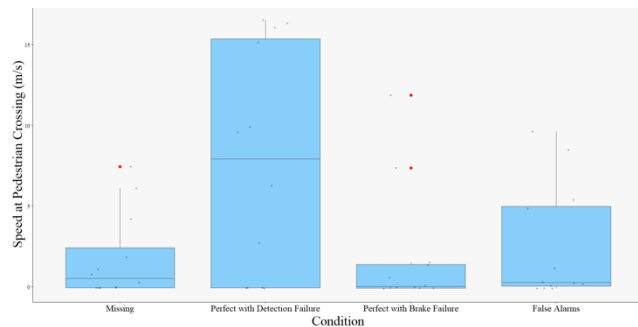


Figure 8. Vehicle speed during pedestrian crossing in the final event

C. Trust in the System

To measure participants' trust in the system, we examined the behavioral demonstration of reliance or non-reliance on the system's behaviors with self-reported measures, using the questionnaire by Jian, Bisantz, and Drury [20]. This

empirical test of trust is similar to the methodology used by Miller et al. [30]. Participants completed the trust questionnaire before and after the simulated driving experience. A repeated-measures ANOVA did not find any statistically significant change between pre-drive and post-drive assessments of the system, nor was there a significant difference in trust scores as a result of the different experimental conditions (see Fig. 9).

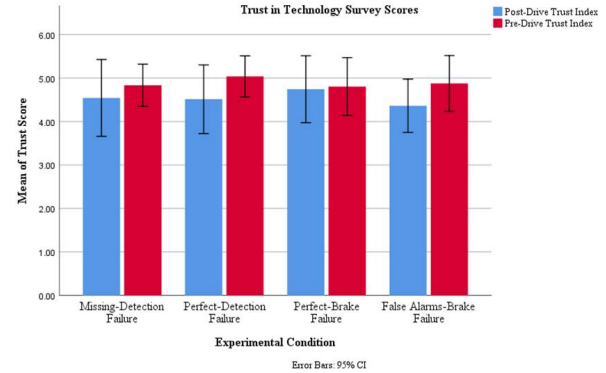


Figure 9. Self-reported measurement of trust in the system

V. DISCUSSION AND CONCLUSIONS

For the final event, participants' driving behavior varied with respect to driving speed when approaching the event, minimum distance from the crossing pedestrians (the person and dog) at the event, and whether or not they collided with the pedestrians. An initial analysis of driving performance under different notification regimes, with particular attention on the final event, suggests that misses for non-fatal events can lead to improved driving performance when an eventual critical event does occur.

In the Missing condition, where the vehicle did not notify drivers or brake for three pedestrians crossing the road, most participants maintained a large minimum distance from the final event pedestrians and did not collide with them. We expect that this driving behavior was motivated by an abundance of caution, in turn caused by low trust in the system, which had exhibited multiple misses.

In contrast, in the Perfect with Detection Failure condition—where the vehicle had correctly notified drivers and stopped for all six preceding pedestrian crossings—most participants did collide with the final event pedestrians, as confirmed by their lower minimum distance from them. These collisions also occurred at relatively high speeds, approaching 16.52 m/s. We expect that this driving behavior was due to complacency resulting from reliance on the automated system, which participants expected to perform perfectly well, having previously experienced only accurate detections and rejections.

In the False Alarms condition, where the vehicle notified drivers and stopped for all events—including those where there was no apparent threat—participants drove at high speed, yet maintained a relatively high minimum distance from the final event pedestrians. The result initially seems counter-intuitive, as most participants in the condition do successfully avoid a collision at the final event. However, when we analyzed the drive videos, we observed that participants *barely* avoided colliding with the pedestrians. In effect, they oftentimes navigated the final event as a near avoidance (see Fig. 10). This behavior may be due to

vigilance levels changing, with participants exhibiting low levels of vigilance throughout the drive, increasing to high vigilance when triggered by the audio alert.

To this latter point, the audio alert likely influenced drivers' vigilance during the final event in both Perfect with Detection Failure and False Alarms conditions. Participants in both conditions successfully navigated the final event, although, as indicated in the drive videos, often with a near avoidance—thereby avoiding the outcome that we anticipated would mirror the townspeople in the Aesop's fable. Audio notifications, regardless of braking behavior, have a significant impact on driving performance when facing hazards that could cause serious injury.



Figure 10. Front and aerial view of road from participant driving in False Alarms condition and exhibiting near avoidance

In conclusion, our study provides a preliminary analysis of the effect of drivers' trust and complacency, along with correlated vigilance, regarding an automated driving system. Our hypothesis comprised two parts: that a system exhibiting misses can result in improved driver response during critical events, and a system with perfect performance and false alarms can lead to complacency that negatively influences driver response. Our findings support the first part, although it is unclear if driving improvement was due to changes in levels of trust and reliance on the system when it performed perfectly.

VI. LIMITATIONS AND FUTURE WORK

One shortcoming of simulator studies is that driver trust in autonomous systems will likely be different in the simulator than on the road with real traffic. The simulator also presents a skewed perception of danger and urgency. However, given the danger of testing misses and false alarms in the real world, with real dangers, simulator studies are an ethical initial compromise. In future studies, we hope to modify the design to allow experiments on physical roads, and thus elicit participant responses that are closer to their real-world behaviors.

We also hope to extend the study duration so that participants are presented with even more false alarms and misses, because we believe such additional training will further alter and change their driver behavior, perhaps causing more significant final event results. A larger participant pool may also further reinforce results.

Additionally, as described in the study design above, the final event occurred on a right-hand curve. While such design provides for an element of surprise in the final event, and has been used in prior studies examining automation failure, it also means that participants might already plan to slow down prior to entering the curve and ahead of the event, thereby making it easier for them to react. Mok et al. [31, 32, 33] studied automation failures and participant responses by situating critical events on curves for similar reasons, although in Mok's studies, the vehicle was in autonomous mode for several minutes prior to encountering the events.

Participants were therefore not as vigilant compared to when they were already driving and needed to react to turns. In future studies, we hope to redesign the final test event such that we account for natural curve braking behavior, in addition to the element of surprise.

Finally, because participants were driving without autonomous control, they may have maintained, as mentioned above, an extra level of vigilance and awareness. That is, while there were no significant differences in trust, this behavior is possibly a result of higher levels of self-trust due to participants navigating the events themselves. We expect results to differ when the automated system is driving. With higher levels of automation, the trust metric may show significant differences between conditions, especially when the participant has little to no control of the vehicle's driving. Reversing the roles of the driver and computer, and tasking the driver to supervise an imperfect automated system, will allow us to further explore the issues presented by partially-automated driving in its current form. In sum, while there has been a great deal of research in this area, further study is required to continue to refine our understanding of how to bias future ADAS to optimize human-system safety and performance.

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